

Odd preimages of powers of two under the Collatz map

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*This note is **not** a proof of the Collatz conjecture.*

Abstract

We prove that an odd positive integer n satisfies $3n + 1 = 2^m$ for some integer $m \geq 1$ if and only if

$$n = n_k = \frac{4^k - 1}{3}$$

for some integer $k \geq 1$. Equivalently, the binary expansion of n is of the form $1(01)^{k-1}$. As a corollary, if a Collatz orbit ever reaches a power of two, then the last odd term before the pure power-of-two tail belongs to the family $\{n_k\}_{k \geq 1}$. We record the standard observation that the Collatz conjecture is equivalent to the statement that every orbit meets this family. The latter claim is not proved here.

1. Definitions

Let $\mathbb{N}$ denote the positive integers. The Collatz map $C: \mathbb{N} \rightarrow \mathbb{N}$ is defined by

$$C(n) = \begin{cases} n/2 & \text{if } n \text{ is even,} \\ 3n + 1 & \text{if } n \text{ is odd.} \end{cases}$$

The orbit of $N \in \mathbb{N}$ is the sequence $N, C(N), C(C(N)), \dots$

For each integer $k \geq 1$ set

$$n_k = \frac{4^k - 1}{3}.$$

The first values are

$$n_1 = 1, n_2 = 5, n_3 = 21, n_4 = 85, n_5 = 341.$$

2. The lemma

Lemma 1.

Let $n \in \mathbb{N}$ be odd. The following conditions are equivalent.

There exists an integer $m \geq 1$ such that $3n + 1 = 2^m$. .1

There exists an integer $k \geq 1$ such that $n = n_k = (4^k - 1)/3$. .2

There exists an integer $k \geq 1$ such that .3

$$n = 1 + 2^2 + 2^4 + \dots + 2^{2k-2},$$

i.e. the binary expansion of n is $1(01)^{k-1}$.

Proof.

We prove $(1) \Rightarrow (2) \Rightarrow (3) \Rightarrow (1)$.

.(1) $\Rightarrow$ (2)

Assume $3n + 1 = 2^m$. Then $2^m \equiv 1 \pmod{3}$. The powers of two modulo 3 alternate:

$$2^{2\ell} \equiv 1 \pmod{3}, 2^{2\ell+1} \equiv 2 \pmod{3}.$$

Hence m is even. Write $m = 2k$ with $k \geq 1$. Then

$$n = \frac{2^{2k} - 1}{3} = \frac{4^k - 1}{3} = n_k.$$

.(2) $\Rightarrow$ (3)

If $n = n_k$, then

$$4^k - 1 = (2^2 - 1)(1 + 2^2 + 2^4 + \dots + 2^{2k-2}),$$

so

$$n_k = 1 + 4 + 4^2 + \dots + 4^{k-1} = 1 + 2^2 + 2^4 + \dots + 2^{2k-2}.$$

In binary this is a 1 in each even bit position and a 0 in each odd bit position, that is $1(01)^{k-1}$.

.(3) $\Rightarrow$ (1)

If the binary expansion of n is $1(01)^{k-1}$, then $n = n_k$, and therefore

$$3n + 1 = 4^k = 2^{2k}.$$

Thus $3n + 1$ is a power of two.

This completes the cycle of equivalences. $\square$

3. Entry into the power-of-two tail

Corollary 2.

Suppose the Collatz orbit of $N \in \mathbb{N}$ contains a power of two 2^m with $m \geq 1$. Then the last odd term occurring in the orbit before the subsequent tail

$$2^m \rightarrow 2^{m-1} \rightarrow \dots \rightarrow 4 \rightarrow 2 \rightarrow 1$$

equals n_k for $k = m/2$ if that odd term exists. In particular, if the orbit reaches a power of two by applying $n \mapsto 3n + 1$ to an odd integer, then the exponent m is even.

Proof.

While a term is even and greater than 1, the map C only divides by 2. Consequently a power of two appears in the orbit either as the starting value itself, or as the image $C(n) = 3n + 1$ of the preceding odd term n . In the second case $3n + 1 = 2^m$, so Lemma 1 gives $n = n_k$ and $m = 2k$. $\square$

4. What is not proved**Remark 3.**

Lemma 1 classifies only those odd integers that land on a power of two in a single step $n \mapsto 3n + 1$. It does not assert that an arbitrary starting value ever reaches the set $\{n_k : k \geq 1\}$.

Proposition 4.

The following statements are equivalent.

For every $N \in \mathbb{N}$ the orbit of N under C eventually reaches 1. .1

For every $N \in \mathbb{N}$ the orbit of N under C eventually reaches a power of two. .2

For every $N \in \mathbb{N}$ the orbit of N under C contains some term n_k . .3

Proof.

If the orbit reaches 1, then it reaches the power of two 2^0 or, more properly in the usual formulation, it reaches $4 = 2^2$ and then 2 and 1. Conversely, once a power of two appears, repeated division by 2 yields 1. This gives the equivalence of (1) and (2). The equivalence of (2) and (3) is Corollary 2 together with the trivial observation that $C(n_k) = 2^{2k}$. $\square$

Statement (1) is the Collatz conjecture. It remains open and is not claimed in this note.

5. Examples

k	n_k	binary	$3n_k + 1$
1	1	1	$4 = 2^2$
2	5	101	$16 = 2^4$
3	21	10101	$64 = 2^6$
4	85	1010101	$256 = 2^8$
5	341	101010101	$1024 = 2^{10}$

In hexadecimal the same numbers produce patterns of the digit 5, not strings of F:

$$21 = 15_{16}, 85 = 55_{16}, 341 = 155_{16}, 1365 = 555_{16}.$$

By contrast, integers of the form $2^m - 1$ have binary expansion $11 \dots 1$ and hexadecimal form $F \dots F$, $1F \dots F$, $3F \dots F$ or $7F \dots F$. Except for trivial small cases they are not the odd preimages classified by Lemma 1.

6. Note on priority and scope

The classification of odd solutions to $3n + 1 = 2^m$ is elementary and should be regarded as standard. The purpose of this note is to record a self-contained proof, to fix the exact binary description $1(01)^{k-1}$, and to separate this unconditional lemma from the Collatz conjecture.